

Практикум D2

156 marks of the archive, ten techniques, twelve tasks. The first practicum in the statistics section, and the first anywhere in the series whose answer is just a number between zero and one.

О практикуме

Первый практикум секции D и первый по статистике вообще. Тема `statistics.probability` — 77 блоков и 237 баллов; D2 берёт 50 блоков и 156 баллов, то есть всё, где ответ получают из устройства вероятностного пространства. Доля была 0.658 — те самые 156 из 237; при сборке D4 она стала 0.675, 160 из 237: в неё вошли 4 балла ноябрьского дубля 2023 года (TZ2 P1 Q2), как дубль входит в доли всех остальных практикумов.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-d2-probability

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

D2 — Probability: events, conditioning, independence, trees

156 marks of the archive, ten techniques, twelve tasks. The first practicum in the statistics section, and the first anywhere in the series whose answer is just a number between zero and one.

That last fact is what makes the topic feel easy and score badly. Nothing here is hard to compute. What is hard is knowing **which space you are computing in** — and every lost mark in this topic is a mark lost to the wrong space, not to the wrong arithmetic.

The one idea

A probability question does not give you a number to manipulate. It gives you a **space**, and the space gives you every number you will ever need from it. Two events cut the space into four cells; a

tree cuts it into paths; three dice cut it into 216 triples. Once the cells are filled in, the question is arithmetic. Before they are, no formula will save you.

So the work, always, is: *what are the outcomes, and what are their weights?* Draw the diagram. Draw the tree. Write the 216 down as a rule. Then read the answer off.

How the checks work

They do not know the answer. `verify_event` takes the conditions your question states — $P(A) = 0.65$, $P(A | B) = \frac{1}{4}$, *A and B are independent* — treats them as equations on the weights of the four cells, solves them, and computes what was asked from the definition. `verify_probability` is handed a space written out as outcomes and weights, checks that the weights come to one, and adds up the ones you want.

When you are wrong they say **how**. The slips in this topic have names, and each

is rebuilt from your own space:

what you wrote	what the check says
$P(B A)$ instead of $P(A B)$	the conditional is the wrong way round
$P(A \cap B)$ instead of $P(A B)$	it has not been divided by the condition yet
$P(A) + P(B)$ for $P(A \cup B)$	the intersection is counted twice
$P(A)P(B)$ for $P(A \cap B)$	the question never promised independence
$1 - p$ instead of p	that is the opposite event
anything above 1	a probability is never above one

Order of work

level	what it means	tasks
	the space is drawn for you; read it	1-3
	build the space, then read it	4-8
	build it, then run it backwards	9-12

Every task is a real past-paper question, cited. The full archive of the topic — all fifty questions — is in the companion notebook *D2 archive: probability*.

62% of these marks carry a calculator, and unusually for this series that is close to honest: about 30 marks out of 156 genuinely need one, because normal and binomial probabilities sit on the branches of the trees in tasks 9 and 10. The other 126 are paper.

```
import sys
sys.path.append('.')          # from
practicum/statistics to practicum/kit/
import sympy as sp            # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
Rational, Eq, events, P(...)

language('en')                # this
notebook is in English, and so are the checks

k_, p_, y_ = symbols('k p y')

print('ready; sympy', sp.__version__)
print('a probability: ', Rational(9, 14))
print('two numbers:    ', [Rational(9, 28),
Rational(5, 14)])
print('events:         ', events('A B'))
```

Map of the ten techniques

#	technique	you recognise it by	it reduces to
1	the algebra of events	a list of $P(\cdot)$ for combinations of two events	four cells that add to one
2	conditional probability	the word <i>given</i> , or a vertical bar	$P(A \cap B)/P(B)$
3	independence	the word <i>independent</i> , in the question or the answer	one more equation: $P(A \cap B) = P(A)P(B)$
4	trees	the experiment has stages	multiply along, add across
5	repeated trials	the same trial over and over	$p(1 - p)^{n-1}$ for the first success
6	total probability	the population splits into parts	$\sum P(C_i)P(E C_i)$
7	Bayes	the effect is known, the cause is asked	one path over technique 6

#	technique	you recognise it by	it reduces to
8	without replacement	nothing goes back	the denominator shrinks — and so does the numerator
9	counting	equally likely outcomes you can count	how many fit / how many there are
10	a letter in the space	an unknown k , p or x inside a probability	an equation, and a root thrown away

Techniques 1-3 all live on the same picture and are really one skill seen three ways. Techniques 6 and 7 are one machine run forwards and backwards. The genuinely separate ones are 4, 5, 8, 9 — the four ways a space gets built — and 10, which is the only place where an answer is rejected for not being a probability.

Part 1. Four cells

Theory: the diagram is the method

Two events A and B cut the sample space into four pieces and no more:

$$A \cap B, \quad A \cap B', \quad A' \cap B, \quad A' \cap B'$$

Their probabilities add to one. **Everything** you can be asked about A and B is a sum of some of them, so filling the four cells in solves the question before you have used a single formula.

The formulas are just readings of the picture:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

— because adding $P(A)$ and $P(B)$ counts the middle cell twice.

$$P(A' \cap B') = 1 - P(A \cup B)$$

— De Morgan: *neither* is the complement of *either*. This is the one that goes wrong. $1 - P(A) - P(B)$ is not it, and the difference is exactly the middle cell.

Conditioning is a new diagram

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

The bar throws away everything outside B and asks the question again in what is left. Two things follow that people get wrong constantly.

The order matters. $P(A | B)$ and $P(B | A)$ have the same numerator and different denominators. They agree only when $P(A) = P(B)$, which is an accident.

The complement is of the event, not of the condition. $P(A' | B) = 1 - P(A | B)$ is true — the events inside B still add to one. But $P(A | B')$ has nothing to do with $P(A | B)$: it is a different question about a different piece of the space.

What the markscheme actually wants. One note in the archive is worth reading twice: *"For the final mark, the value must be stated as the candidate's answer, or labeled as $P(A' \cap B')$ in their Venn diagram. Just seeing an unlabeled number in the*

correct region earns M1A0." A number in the right place on a picture is not an answer until you say what it is.

Task 1 ● — **November 2023 TZ1**
Paper 1 Q2, 4 marks

Events A and B are such that $P(A) = 0.65$, $P(B) = 0.75$ and $P(A \cap B) = 0.6$.

(a) Find $P(A \cup B)$. **(b)** Hence, or otherwise, find $P(A' \cap B')$.

```
A, B = events('A B')
given = [(P(A), 0.65), (P(B), 0.75), (P(A &
B), 0.6)]

qla = ...
qlb = ...

verify_event('1a', qla, given, P(A | B))
verify_event('1b', qlb, given, P(~A & ~B))
```

Task 2 ● — **May 2021 TZ2 Paper 2**
Q3(a)(b), 4 marks

At a school, 70% of the students play a sport and 20% of the students are involved in theatre. 18% of the students do neither activity. A student is selected at random.

- (a)** Find the probability that the student plays a sport and is involved in theatre.
- (b)** Find the probability that the student is involved in theatre, but does not play a sport.

The 18% is the fourth cell, and it is the only reason this question is solvable. Percentages in words are probabilities; put them on the diagram as they arrive.

```
S, T = events('S T')
school = [(P(S), Rational(7, 10)), (P(T),
Rational(2, 10)),
          (P(~S & ~T), Rational(18, 100))]

q2a = ...
q2b = ...

verify_event('2a', q2a, school, P(S & T))
verify_event('2b', q2b, school, P(T & ~S))
```

Task 3 — **November 2025 TZ3** **Paper 1 Q2, 6 marks**

Events A and B are such that $P(A) = \frac{1}{2}$,
 $P(B) = \frac{1}{3}$ and $P(A | B) = \frac{1}{4}$.

- (a)** Find the probability that both events A and B occur. **(b)** Find $P(A | B')$.

Part (b) is the trap named above. $1 - \frac{1}{4} = \frac{3}{4}$ is not the answer, and it is not close to the answer.

```
A, B = events('A B')
given = [(P(A), Rational(1, 2)), (P(B),
Rational(1, 3)),
        (P(A, given=B), Rational(1, 4))]
```

q3a = ...
q3b = ...

```
verify_event('3a', q3a, given, P(A & B))
verify_event('3b', q3b, given, P(A,
given=~B))
```

Part 2. Building the space

Theory: four ways a space gets made

Part 1 handed you a filled diagram. From here you have to draw it.

Independence is an equation, not an adjective

$$\begin{aligned} A \text{ and } B \text{ independent} &\iff P(A \cap B) \\ &= P(A) P(B) \end{aligned}$$

That is a **definition**, and it works in both directions, which is the whole point.

- The question **asks** whether they are independent $\rightarrow$ compute both sides and compare. The verdict alone earns nothing: the markscheme says *"Both conclusion and reasoning are required. Do not split the A2."*
- The question **says** they are independent $\rightarrow$ you have gained an equation. Usually it is the one that

makes the problem solvable, and usually it is quadratic, which is why those parts are worth four marks and not two.

And when it is quadratic, **one root goes in the bin** — not because it is extraneous, but because it is bigger than one and probabilities are not. Say so in writing. A typical such quadratic offers, say, $P(B) = 0.4$ and $P(B) = \frac{9}{5}$, and the markscheme awards **A1 instead of A2** to anyone who hands in both: the second is greater than one and is therefore not a probability at all.

Trees: along a branch multiply, between branches add

The first level carries the probabilities of the first stage; every node of the second level carries **conditional** probabilities. Each fan of branches adds to one — check that, always, because it is free and it catches everything.

Write the branch complement as $1 - p$ and define it. May 2023 is explicit:

"award A0 for G' branch labelled q unless explicitly defined as $1 - p$."

Repeated trials: the first success

$$P(\text{first success on trial } n) = p(1 - p)^{n-1}$$

$n - 1$ failures, then a success. The exponent is $n - 1$; that off-by-one is the entire difficulty. And "at least one in n " is always $1 - (1 - p)^n$ — never a sum, because the sum is where a term gets dropped.

A letter in the space

An unknown k , p or x sits inside the probabilities and a probability computed from it is given. You do not substitute; you write the equation and solve it. Then you throw a root away and **say why** — a whole R1 mark hangs on one short sentence naming the reason, and "greater than one" is the whole of it.

Task 4 — May 2025 TZ2 Paper 1 Q4, 7 marks

Events A and B are such that $P(A \cup B) = \frac{5}{8}$ and $P(A \cap B') = \frac{7}{24}$.

(a) Find $P(B)$. **(b)** Given that events A and B are independent, find $P(A' | B)$.

Two numbers do not fill four cells, and they do not have to: $P(B)$ is determined anyway, because $A \cup B$ splits into $A \cap B'$ and B with no overlap. Part (b) is where independence earns its keep.

```
A, B = events('A B')
given = [(P(A | B), Rational(5, 8)), (P(A &
~B), Rational(7, 24))]

q4a = ...
q4b = ...

verify_event('4a', q4a, given, P(B))
verify_event('4b', q4b, given + [Eq(P(A & B),
P(A)*P(B))], P(~A, given=B))
```

Task 5 — May 2022 TZ2 Paper 2 Q3, 6 marks

Events A and B are independent and $P(A) = 3P(B)$.

Given that $P(A \cup B) = 0.68$, find $P(B)$.

Independence turns the addition rule into a quadratic. Solve it, then look at both roots and decide out loud which one is a probability.

```
A, B = events('A B')
given = [Eq(P(A & B), P(A)*P(B)),
         Eq(sympify(P(A)), 3*sympify(P(B))),
         (P(A | B), Rational(68, 100))]

q5 = ...

verify_event('5', q5, given, P(B))
```

Task 6 — **November 2021 Paper 1 Q4, 5 marks**

Box 1 contains 5 red balls and 2 white balls. Box 2 contains 4 red balls and 3 white balls.

(a) A box is chosen at random and a ball is drawn. Find the probability that the ball is red.

Let A be the event that "box 1 is chosen" and R be the event that "a red ball is drawn".

(b) Determine whether events A and R are independent.

For (b) hand over the two numbers the comparison rests on, in the order $P(A)P(R)$ then $P(A \cap R)$ — which is what the markscheme wants to see under the conclusion.

```
boxes = {('1', 'red'): Rational(1,
2)*Rational(5, 7),
         ('1', 'white'): Rational(1,
2)*Rational(2, 7),
         ('2', 'red'): Rational(1,
2)*Rational(4, 7),
         ('2', 'white'): Rational(1,
2)*Rational(3, 7)}
A, Rd = events('A R')
tree = [(P(A), Rational(1, 2)), (P(Rd,
given=A), Rational(5, 7)),
        (P(Rd, given=~A), Rational(4, 7))]
```



```
q6a = ...
q6b = ...          # [P(A)*P(R), P(A and R)]
```



```
verify_probability('6a', q6a, boxes, lambda
o: o[1] == 'red')
verify_independence('6b', q6b, tree, A, Rd)
```

Task 7 🟡 — *May 2024 TZ1 Paper 3*
Q1(a), 5 marks

In a new computer game, each time a player performs an action there is a random chance that the action will be *boosted*. In the first model the probability that an action will be boosted is constant. Suppose that probability is 0.1.

(a) Find the probability that the first boost occurs on the third action. **(b)** Find the probability that at least one boost occurs in the first six actions.

Part (b) through the complement. Six terms added up give the same number and take six times as long, and the sixth is the one that gets dropped.

```

boost = {n: Rational(1, 10)*Rational(9, 10)**
(n - 1) for n in range(1, 400)}
boost['never'] = Rational(9, 10)**399

q7a = ...
q7b = ...

verify_probability('7a', q7a, boost, lambda
o: o == 3)
verify_probability('7b', q7b, boost, lambda
o: o != 'never' and o <= 6)

```

Task 8 — May 2024 TZ2 Paper 1 Q4(b), 4 marks

A species of bird nests in Spring with probability k and in Summer with probability $\frac{k}{2}$, in each season independently of the other. It is known that the probability of not nesting in Spring **and** not nesting in Summer is $\frac{5}{9}$.

(i) Show that $9k^2 - 27k + 8 = 0$. **(ii)** Both $k = \frac{1}{3}$ and $k = \frac{8}{3}$ satisfy that equation. State why $k = \frac{1}{3}$ is the only valid solution.

Hand over k ; the check puts it back into the sentence the question states, which is (i) done properly. Part (ii) is one line of

prose and one whole mark — write it before you look at the solution.

```
q8 = ...                # [k]

verify_constants('8', q8, [k_], [
    ('not nesting in Spring and not nesting
in Summer has probability 5/9',
    Eq((1 - k_)*(1 - k_/2), Rational(5,
9))),
], domain=Interval(0, 1))
```

Part 3. Running it backwards

Theory: total probability, Bayes, and two ways of counting

Down the tree: total probability

$$P(E) = \sum_i P(C_i) P(E | C_i)$$

The population splits into parts — two machines, three clerks, two kinds of apple — and each part contributes **its share times what happens inside it**. Check first that the parts cover everything and do not overlap; that is what makes the sum legitimate.

The standard wrong answer is to add the conditional probabilities without their weights. Three machines faulty 2%, 5% and 9% of the time give $0.02 + 0.05 + 0.09 = 0.16$ — but every one of the three rates is below 0.09, so the true answer cannot be above it either. A weighted

average always lands between the smallest and the largest of the things being averaged; if yours does not, you dropped the weights.

Back up the tree: Bayes

$$P(C | E) = \frac{P(C) P(E | C)}{P(E)}$$

The tree runs cause $\rightarrow$ effect; the question runs effect $\rightarrow$ cause. The numerator is **one path**; the denominator is the whole of $P(E)$, which is the sum you just did. There is nothing else to it: the hard part was total probability, and Bayes is a division.

The markscheme wants the conditional written **in context** — *"not just $P(A | B)$ ".* Letters earn nothing; name the events.

Bayes is also where intuition is worst, and how far the conditioning moves the answer depends entirely on how far the rates differ. If one source handles most of the work and its error rate is close to the average, the share of the errors it

accounts for is close to its share of the work — the conditioning barely shifts anything. If instead one group is rare but many times more likely to produce the effect, the shift is enormous, and a majority before the evidence can become a minority after it. Before computing, ask which of the two situations you are in; it tells you what kind of answer to expect.

Without replacement

$$P(YY) = \frac{y}{r+y} \cdot \frac{y-1}{r+y-1}$$

Both the numerator and the denominator drop by one. Everyone remembers the denominator; the numerator is the one that gets left alone.

Counting an equally likely space

When outcomes are equally likely and countable, probability is $\frac{\text{how many fit}}{\text{how many there are}}$, and all the work is in the counting. Count **by cases**, never by listing, and count **ordered tuples**:

rolling 1, 2, 3 is a different outcome from 3, 2, 1, and both are among the 216.

Task 9 ● — May 2025 TZ3 Paper 2
Q11(c), 6 marks

Amanda, Bryce and Carmen enter data from surveys into a database. Surveys entered by Amanda, Bryce and Carmen are inaccurate 8%, 6% and 11% of the time respectively. From the surveys assigned to the three of them, Amanda enters 55%, Bryce 25% and Carmen 20%.

Find the probability that a randomly selected survey was

(i) entered inaccurately; **(ii)** entered by Amanda, given that the survey was entered inaccurately.

```

clerk = {'Amanda': (Rational(55, 100),
Rational(8, 100)),
        'Bryce': (Rational(25, 100),
Rational(6, 100)),
        'Carmen': (Rational(20, 100),
Rational(11, 100))}
survey = {(who, ok): share*(bad if ok ==
'wrong' else 1 - bad)
          for who, (share, bad) in
clerk.items() for ok in ('wrong', 'right')}

q9a = ...
q9b = ...

verify_probability('9a', q9a, survey, lambda
o: o[1] == 'wrong')
verify_probability('9b', q9b, survey,
                  lambda o: o[0] ==
'Amanda', given=lambda o: o[1] == 'wrong')

```

Task 10 — **November 2025 TZ1** **Paper 2 Q7, 7 marks**

Two machines are used in a factory to manufacture semiconductors. Machine A manufactures defective semiconductors 10% of the time, while machine B manufactures defective semiconductors 5% of the time. A randomly selected machine manufactures ten semiconductors. Both machines are equally likely to be selected.

(a) Find the probability that exactly two semiconductors are defective. **(b)** Given that exactly two semiconductors are defective, find the probability that they were manufactured by machine A.

The branch probabilities are binomial — that is D3's business and the GDC's. What is D2's business is the tree they sit in and the direction the question runs.

```
chip = {(name, d): Rational(1,
2)*binomial(10, d)*q**d*(1 - q)**(10 - d)
        for name, q in (('A', Rational(1,
10))), ('B', Rational(1, 20)))
        for d in range(11)}

q10a = ...
q10b = ...

verify_probability('10a', q10a, chip, lambda
o: o[1] == 2)
verify_probability('10b', q10b, chip,
                    lambda o: o[0] == 'A',
given=lambda o: o[1] == 2)
```

Task 11 — **May 2023 TZ2 Paper 2 Q11(d)(e), 8 marks**

A game of chance involves drawing three balls out of a box without replacement.

The box initially contains 10 red balls and y yellow balls. Let $P(YYY)$ represent the probability of drawing three yellow balls without replacement.

(a) Find an expression for $P(YYY)$ in terms of y .

A yellow ball is added, so the box now contains 10 red balls and $(y + 1)$ yellow balls. The probability of drawing three yellow balls is now twice the probability in part (a).

(b) Find the initial number of yellow balls in the box.

```

q11a = ...           # in terms of y_
q11b = ...           # the number of yellow
balls the box started with

# the model stage by stage: at draw i there
are y - i yellow left of y + 10 - i
stages = [(y_ - i)/(y_ + 10 - i) for i in
range(3)]
verify_identity('11a', q11a, Mul(*stages),
var=y_, samples=(4, 7, 11, 20))

verify_constants('11b', [q11b], [y_], [
    ('adding one yellow ball doubles the
probability of drawing three',
    Eq((y_ + 1)*y_*(y_ - 1)/((y_ + 11)*(y_ +
10)*(y_ + 9)),
        2*y_*(y_ - 1)*(y_ - 2)/((y_ + 10)*(y_
+ 9)*(y_ + 8))))),
    ])

```

Task 12 — May 2024 TZ2 Paper 3 Q2(c)(f), 12 marks

Consider quadratic functions $f(x) = ax^2 + bx + c$ whose coefficients a , b and c are randomly generated in turn by rolling an unbiased six-sided die three times.

(a) By considering the discriminant, or otherwise, show that the probability of the graph having only one x -intercept is $\frac{5}{216}$.

(b) Let p be the probability of the graph having two **distinct** x -intercepts. Find the value of p .

Twelve marks of counting. Part (b) is done by cases on the value of ac , and two of those cases are where everyone loses it: at $ac = 4$ the discriminant is exactly zero when $b = 4$ — that belongs to part (a) — and at $ac = 6$ there are four ordered pairs, not two.

```
rolls = {(u, v, w): Rational(1, 216)
          for u in range(1, 7) for v in
range(1, 7) for w in range(1, 7)}

q12a = ...
q12b = ...

verify_probability('12a', q12a, rolls, lambda
t: t[1]**2 - 4*t[0]*t[2] == 0)
verify_probability('12b', q12b, rolls, lambda
t: t[1]**2 - 4*t[0]*t[2] > 0)
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you**

would make first. That is the decision this topic turns on, and on the paper you get about five seconds of it.

code	technique
venn	four cells: addition rule, complement, De Morgan
given	conditional probability
indep	independence, as a test or as an extra equation
tree	stages: multiply along, add across
first	repeated trials, the first success
total	the population splits into parts
bayes	the effect is known, the cause is asked
noreplace	nothing goes back in the box
count	equally likely outcomes, counted
letter	an unknown inside the probability

1. $P(A) = 0.65$, $P(B) = 0.75$, $P(A \cap B) = 0.6$. Find $P(A' \cap B')$.
2. $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A | B) = \frac{1}{4}$. Find $P(A | B')$.
3. Given that a randomly selected muffin weighs less than 61 g, find the probability that it is chocolate.

4. $P(A \cap B') = 0.16$ and $P(A' \cap B) = 0.36$; given $P(A \cap B) = x$, find x .
5. Show that the probability of a randomly generated quadratic having only one x -intercept is $\frac{5}{216}$.
6. A box is chosen at random and a ball is drawn. Find the probability that the ball is red.
7. It is known that the probability of not nesting in either season is $\frac{5}{9}$. Show that $9k^2 - 27k + 8 = 0$.
8. Find an expression for $P(YYY)$ in terms of y , where the box holds 10 red and y yellow.
9. Find the probability that a randomly selected survey was entered inaccurately.
0. Find the probability that the first day that it rains in May is on the 10th day.
1. An employee is selected at random from the group who work more than 40 hours. Find the probability that they work less than 55 hours.

2. Determine whether events A and R are independent.

```
answers = {
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
    7: '', 8: '', 9: '', 10: '', 11: '', 12:
    '',
}

trigger_check(answers, {1: '29901ef03112', 2:
'5b729e0f6197', 3: '110f945a922b', 4:
'1e76c19f7fab', 5: '6c35493a2b93', 6:
'dc9c5edb8b2d', 7: 'f80105efab6d', 8:
'a3c6e6ae1c64', 9: '11239872d178', 10:
'a7937b64b8ca', 11: '5b729e0f6197', 12:
'1e76c19f7fab'})
```

On the timer

*May 2022 TZ1 Paper 2 Q5 — 6 marks.
Eight minutes.*

Top of the notebook covered, paper and pen, formula booklet allowed.

Let A and B be two independent events such that $P(A \cap B') = 0.16$ and $P(A' \cap B) = 0.36$.

(a) Given that $P(A \cap B) = x$, find the value of x .

(b) Find $P(A' \mid B')$.

date	time	notes

```
A, B = events('A B')
paper = [Eq(P(A & B), P(A)*P(B)),
         (P(A & ~B), Rational(16, 100)),
         (P(~A & B), Rational(36, 100))]
```

```
qt_a = ...
qt_b = ...
```

```
verify_event('timer (a)', qt_a, paper, P(A &
B))
verify_event('timer (b)', qt_b, paper, P(~A,
given=~B))
```



Solutions

1 (a) $P(A \cup B) = 0.65 + 0.75 - 0.6 = \boxed{0.8}$.

The subtraction is there because adding $P(A)$ and $P(B)$ counts the middle cell twice.

1 (b) De Morgan: $A' \cap B' = (A \cup B)'$, so $1 - 0.8 = \boxed{0.2}$.

Not $1 - 0.65 - 0.75$, which is negative. If your answer comes out negative you have used $1 - P(A) - P(B)$; the check will refuse it before you notice.

2 (a) The four cells add to one:

$$P(S) + P(T) + P(S' \cap T') - P(S \cap T) = 1$$

$$0.7 + 0.2 + 0.18 - P(S \cap T) = 1$$

$$\implies P(S \cap T)$$

$$= \boxed{0.08}$$

Or: $P(S \cup T) = 1 - 0.18 = 0.82$, then $0.82 = 0.7 + 0.2 - P(S \cap T)$.

2 (b) $P(T \cap S') = P(T) - P(T \cap S) = 0.2 - 0.08 = \boxed{0.12}$.

The markscheme's note on the diagram route is worth reading: *"To obtain the M1 for the Venn diagram all labels must be correct and in the correct sections. For example, do not accept 0.7 in the area corresponding to $S \cap T'$."* $P(S)$ is not the left cell — it is the left cell plus the middle.

3 (a) $P(A \cap B) = P(A | B) P(B) = \frac{1}{4} \cdot \frac{1}{3} = \boxed{\frac{1}{12}}$.

The conditional multiplies the probability of **its own condition**. That is the direction to memorise; the other one gives $\frac{1}{8}$ and is wrong.

3 (b) $P(B') = \frac{2}{3}$, and

$$\begin{aligned} P(A \cap B') &= P(A) - P(A \cap B) \\ &= \frac{1}{2} - \frac{1}{12} \\ &= \frac{5}{12} \end{aligned}$$

$$P(A \mid B') = \frac{5/12}{2/3} = \boxed{\frac{5}{8}}$$

Not $1 - \frac{1}{4} = \frac{3}{4}$. Complementing the **condition** changes which piece of the space you are standing in; complementing the **event** does not. Only the second one is a subtraction from 1.

4 (a) $A \cup B$ is $A \cap B'$ together with B , and those do not overlap:

$$\frac{5}{8} = \frac{7}{24} + P(B) \implies P(B) = \frac{15-7}{24} = \boxed{\frac{1}{3}}$$

Two numbers and four cells, and the answer comes out anyway. It is worth seeing why: $P(A \cap B)$ is genuinely undetermined here, and $P(B)$ does not depend on it.

4 (b) Independence gives $P(A \cap B') = P(A)P(B')$, so $\frac{7}{24} = P(A) \cdot \frac{2}{3}$ and $P(A) = \frac{7}{16}$. Independence again makes the condition irrelevant:

$$P(A' \mid B) = P(A') = 1 - \frac{7}{16} = \boxed{\frac{9}{16}}$$

That last step is the point of independence: conditioning on B tells you nothing about A , so the bar can be dropped. If you did it the long way you got the same number and spent three more minutes.

5 Independence turns the addition rule into a quadratic in $b = P(B)$:

$$0.68 = 3b + b - 3b^2 \implies 3b^2 - 4b + 0.68 = 0$$

$$\begin{aligned} b &= \frac{4 \pm \sqrt{16 - 8.16}}{6} \\ &= \frac{4 \pm 2.8}{6} \\ &\implies b \\ &= 0.2 \text{ or } b \\ &= \frac{17}{15} \end{aligned}$$

$\frac{17}{15} = 1.133\dots > 1$, and a probability is never above one, so $P(B) = \boxed{0.2}$.

The markscheme: *"Award A1 if both answers are given as final answers for $P(B)$."* Two of the six marks are for making the choice and knowing why.

6 (a) $P(R) = P(R \cap B_1) + P(R \cap B_2) = \frac{1}{2} \cdot \frac{5}{7} + \frac{1}{2} \cdot \frac{4}{7} = \frac{9}{14} = \boxed{\frac{9}{14}}.$

6 (b) $P(A) = \frac{1}{2}$ and $P(A \cap R) = \frac{1}{2} \cdot \frac{5}{7} = \frac{5}{14},$
so

$$\begin{aligned} P(A) \\ P(R) &= \frac{1}{2} \cdot \frac{9}{14} = \frac{9}{28} = \frac{9}{28}, \\ P(A \cap R) &= \frac{5}{14} = \frac{10}{28} \end{aligned}$$

They differ, so A and R are **not** independent. $\boxed{[\frac{9}{28}, \frac{5}{14}]}$

The markscheme also accepts the prose version — "*different number of red balls in each box*" — because that is the same statement: which box you are in changes the chance of red, and that is exactly what dependence means.

7 (a) Two misses then a hit: $0.9^2 \times 0.1 = \boxed{0.081}.$

7 (b) Through the complement: $1 - 0.9^6 = 1 - 0.531441 = \boxed{0.469}.$

8 (i) Not nesting in Spring is $1 - k$; not nesting in Summer is $1 - \frac{k}{2}$, and the seasons are independent, so

$$(1 - k) \left(1 - \frac{k}{2} \right) = \frac{5}{9} \implies 1 - \frac{3k}{2} + \frac{k^2}{2} = \frac{5}{9}$$

Multiply by 18:

$$18 - 27k + 9k^2 = 10$$

$$\implies \boxed{9k^2 - 27k + 8 = 0}$$

8 (ii) $k = \frac{8}{3} > 1$, and k is a probability, so it cannot exceed one. Hence $k = \boxed{\frac{1}{3}}$.

The markscheme accepts *"any valid reasoning indicating that any probability cannot be greater than 1 and/or probability cannot be less than 0"*. One sentence, one mark, and it is the mark most often missing.

9 (i) Weighted by the shares:

$$\begin{aligned}
& 0.55(0.08) \\
& + 0.25(0.06) \\
& + 0.20(0.11) = 0.044 \\
& + 0.015 \\
& + 0.022 \\
& = \boxed{0.081}
\end{aligned}$$

Check the shares add to one before you start; here $0.55 + 0.25 + 0.20 = 1$, so the three parts really do cover everybody.

9 (ii)

$$\begin{aligned}
P(\text{Amanda} \mid \text{inaccurate}) &= \frac{0.55 \times 0.08}{0.081} \\
&= \frac{0.044}{0.081} \\
&= \boxed{0.543}
\end{aligned}$$

Amanda enters 55% of the surveys and produces 54% of the errors. Almost no shift — because her error rate, 8%, is close to the overall 8.1%. Conditioning only moves things when the rates differ.

10 (a) Each machine gives a binomial: $B(10, 0.1)$ and $B(10, 0.05)$.

$$P(2 \text{ defective}) = 0.5 \times 0.193710 + 0.5 \times 0.0746347 \\ = \boxed{0.134}$$

10 (b)

$$P(A \mid 2 \text{ defective}) = \frac{0.5 \times 0.193710}{0.134172} \\ = \boxed{0.722}$$

Two defective out of ten is more like machine A than machine B, so seeing it pushes the odds from 50:50 to about 72:28. That is what Bayes is for.

The markscheme's note is worth keeping: *"recognition must be shown in context, either in words or symbols, not just $P(A \mid B)$."*

11 (a) Three yellows drawn from y yellow and 10 red, nothing replaced:

$$P(YYY) = \boxed{\frac{y(y-1)(y-2)}{(y+10)(y+9)(y+8)}}$$

Both parts of every fraction go down by one. Writing $\frac{y}{y+10}$ three times is the

standard wrong answer and it is the with-replacement model.

11 (b) With one more yellow the box holds $y + 1$ yellow and 10 red:

$$\begin{aligned} & \frac{(y+1)y(y-1)}{(y+11)(y+10)(y+9)} \\ &= 2 \cdot \frac{y(y-1)(y-2)}{(y+10)(y+9)(y+8)} \end{aligned}$$

Cancel $y(y-1)$ — legitimate, since $y \geq 3$ — and $(y+10)(y+9)$:

$$\begin{aligned} \frac{y+1}{y+11} &= \frac{2(y-2)}{y+8} \\ &\implies (y+1)(y+8) \\ &= 2(y-2)(y+11) \\ y^2 + 9y + 8 &= 2y^2 + 18y - 44 \\ &\implies y^2 + 9y - 52 \\ &= 0 \\ &\implies (y-4)(y+13) \\ &= 0 \end{aligned}$$

$$\boxed{y = 4}$$

Check: $\frac{4 \cdot 3 \cdot 2}{14 \cdot 13 \cdot 12} = \frac{1}{91}$ and $\frac{5 \cdot 4 \cdot 3}{15 \cdot 14 \cdot 13} = \frac{2}{91}$. Twice, as promised.

12 (a) One x -intercept means $b^2 = 4ac$. Since $4ac$ is a multiple of four, b must be even.

- $b = 2$: $ac = 1$, so $(a, c) = (1, 1)$ — 1 way.
- $b = 4$: $ac = 4$, so $(1, 4), (2, 2), (4, 1)$ — 3 ways.
- $b = 6$: $ac = 9$, so $(3, 3)$ — 1 way, since $(1, 9)$ needs a nine.

Five triples of 216: $\boxed{\frac{5}{216}}$.

12 (b) Two distinct intercepts means $b > 2\sqrt{ac}$, so only $ac \leq 8$ can contribute:

ac	$2\sqrt{ac}$	(a, c) pairs	b that work	product
1	2	1	4	4
2	2.83	2	4	8
3	3.46	2	3	6
4	4	3	2	6
5	4.47	2	2	4
6	4.90	4	2	8
8	5.66	2	1	2

$$4 + 8 + 6 + 6 + 4 + 8 + 2 = 38, \text{ so } p = \frac{38}{216} =$$

$$\boxed{\frac{19}{108}} \approx 0.176.$$

The two rows that go wrong: at $ac = 4$ the bar is exactly 4, and $b = 4$ gives the repeated root that belongs to part (a); at $ac = 6$ there are four ordered pairs, because $6 = 1 \times 6 = 2 \times 3$ and both orders count.

Timer

(a) Write $x = P(A \cap B)$. Then $P(A) = x + 0.16$ and $P(B) = x + 0.36$, and independence says

$$\begin{aligned}x &= (x + 0.16)(x + 0.36) \\&\implies x^2 - 0.48x + 0.0576 \\&= 0 \\&\implies (x - 0.24)^2 \\&= 0\end{aligned}$$

so $x = \boxed{0.24}$ — a repeated root, so no choice to make here.

(b) A and B independent makes A' and B' independent too, so

$$\begin{aligned}
P(A' \mid B') &= P(A') \\
&= 1 - (0.16 + 0.24) \\
&= \boxed{0.6}
\end{aligned}$$

The long way also works: $P(A' \cap B') = 1 - 0.16 - 0.24 - 0.36 = 0.24$ and $P(B') = 1 - 0.6 = 0.4$, giving $0.24/0.4 = 0.6$.